

ActInf GuestStream #010.1 ~ Philip Gerrans

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hello everyone welcome to the active inference lab today it's september 9th for guest stream number 10.1 double digits or double digits with a decimal and we're really honored and appreciative to have professor philip guerins today coming in from australia and also dave joining us on stream we're gonna have a presentation from philip and then we're gonna have some time for discussion and dialogue so if you're watching live please feel free to leave questions in the live chat and we will address those on the stream and we're just really looking forward to this and again appreciative that you're coming on our stream so thanks and take it away phillip okay great uh thank you very much thank you very much for the invitation i spoke to anna kanika who told me that she had a great experience with you guys so i'm very very happy to be able to present and to tell you a little bit about the background to this talk i was originally a philosopher of mind and of cognitive science and i'm very interested in uh the use of models and techniques in cognitive science to understand psychiatric disorder and i found predictive processing very helpful when i was writing a book on this a few years ago but looking back at um some other psychiatric disorders like the cotard delusion and so on i think it's really great to kind of go a little bit deeper and more more fundamental if you think of predictive processing as kind of like a computational structure at the level of representation and algorithm but of course a lot of psychiatric disorders i think you've got a very sort of a bottom-up genesis we really need to know about how the brain is working to control the body and even how the body is working in its social world so the active inference framework for me has been very helpful in rethinking this in this book that i'm writing right now and also i think it's it's actually crucial to understanding a phenomenon that i'm talking about right now that the kotar delusion the cotard delusion is so interesting because um well it's interesting in itself as you'll see it's fascinating phenomenon but it points us to some really interesting uh interesting facts or yeah interesting evidence about the way the mind is mind is working at a really basic level and how that ramifies and cascades upwards throughout cognition and behavior so let's go i'll i'll just tell you what the kotar delusion is it's a delusion of sometimes people think of it in the folk world as the delusion that you're dead i was discovered in 1882

but we described it more accurately i think as a delusion of inexistence so here's a here's someone who has it she had the experience of having no identity or self and being only a body without content she convinced that her brain had disappeared and the whole body was translucent another one more recent one i speak and breathe and eat but i'm dead okay so there's some classic examples but here is a really recent one published in the last couple of years this is fascinating cases coming from mexico there's a really good group there that works on this so mr a was referred to the national institute of neurology because of psychotic and catatonic symptoms his appetite diminished he barely slept he stayed withdrawn and silent he complained about his body saying his tendons had dried up and his organs were getting bigger speak of an internal hemorrhage his organs have been torn apart he began to insist that he was dead i can't feel the blood flowing through my veins didn't deprive him of looking for sharp objects to cut his neck and forearms then he gradually improved first he became mute began to urinate on his clothes refused to eat and then speak finally you know after some treatment he said he's improved slightly but he said i don't have feelings because i'm dead he had a right-sided heart and that it stopped and then he kept on saying even after persisting treatment his nihilistic delusions as they're accurately called you know delusions of inexistence of nothingness i am dead among other human beings it's like a program where i'm dead and he said he was a dead person among the corpses under the earthquake debris now the genesis of this case of qatar delusion is um encephalitis a form of encephalitis where it's an autoimmune reaction against proteins in the brain so it's one of the most basic things that can go wrong at the cellular level and if the protein is in methyl the aspartate the condition is called nmdar antibody encephalitis okay so that's what mr a had but of course our interesting question is if we know what's happening you know at the molecular level at the level of kind of membranes and receptors why on earth would that make you think that you didn't exist so they were addressing that they were in the in the field of addressing the question you know how does the brain enable experience and cognition that's that's the questions of cognitive neuroscience and psychology and anyone else who wants to get involved in that topic phenomenology philosophy okay but it's a really uh pertinent question so here's the outline of the talk i just told you about the delusion and then i'm going to tell you about some of the classic attempts to explain it uh just run through those and point to a problem with those and then in discussing uh one of the most recent and persuasive and really interesting um explanations uh that we owe to alexandre bien i'm going to introduce the active inference and i'm going to show how the active inference approach can rectify a problem with beyonce's account but also a point to

point to some other deep sort of unifying themes across cognitive science and across disorders so the active inference framework i think gives a elegant and persuasive um way to explain not just you know cases like mr a but a lot of other dis other disorders and pathologies where our experience of our self goes astray so things like pain and symbolia where people say that they're experiencing pain but also that their body is experiencing pain but it's not happening to them but it feels as though it's not their pain or depersonalization disorders in which people will say not just that pain is not theirs but they feel outside of or detached from their whole experience okay so and the solution that i'll propose i'll give you um just a short summary right now is that as part of the um project or process of of active inference the body the brain makes uh hierarchical models of itself and the world of deep the deep structure of the world as as populated with objects and properties that the external world with which it interacts but of course the brain also has to make a model of its own internal uh uh functioning the processes that that keep it keep keep us alive and functioning and in order to do that it makes a model of itself as a unified persisting entity and that's what we feel the self that's what we feel when we feel like like persisting people when we feel like ourselves we we are the model that's what we're experiencing activity in the circuitry that makes that uh self-model okay and in a cotard delusion there's the self-model uh goes wrong in a certain way or that process of self-modeling goes wrong in a certain way but we can see other disorders in which that process of self-modelling goes wrong and i think that theory makes it maps really nicely to the um the neural substrates of all these disorders it explains the symptoms and so on so let's go so of course as we saw um such an interesting disorder the the quartile delusion because you know there's there's questions in sort of philosophy and about rational belief fixation and so on i mean how could someone coherently say i've died or i don't exist and there's a question about rationality there and the nature of reasoning and so on on the basis of experience which is interesting in itself which is that's part of the philosophy of delusions i guess or the philosophy of psychiatric disorder where i came into this topic a while ago but of course it's also very interesting in okay how can you also have the experience of not existing you know so it points us to the question of what is going on when you actually feel like a self okay and of course it's also a great test case for theories that try and explain experience and cognition in terms of underlying neural functioning so here's the kind of broad brush of the kind of classic approaches to the these topics that you get in theories of delusion since i guess the early 90s which tried to wed philosophy of belief rational belief fixation to the neuroscience of dilution formation and that the most top-level abstract way of putting it is there's two stages in delusion

formation the first one you know is an unusual binomial as they say sensory experience so something goes wrong you know at the level of sensory motor gives you an unusual experience and then the second stage is a metacognitive response to that experience so this is why someone for example who lost control or was unable to feel unable to feel that they are the agent of their actions when they're picking up things and moving around the world okay that's the loss of it as they call sense of agency due to um you know normally problems in the parietal lobule inferior parietal lobule something goes wrong there with the kind of sensory motor in action of agency and then the people what person will say to explain that weird experience of moving around the world but not feeling like they are the person who initiated the movement or controlling it and say someone else is controlling me you know so delusions of control as they say that would be a classic case we have an anomalous experience at the first stage and then the delusion is a belief or some kind of metacognitive rationalization or story that the person tells okay there's two implications of the um the two-stage theory one is that well if the patient doesn't have a problem with higher order processing then she won't develop a delusion because the delusion is a metacognitive response right so in that case the person should have an anomalous experience she should feel like she's making movements that aren't hers for example or you might feel like you're hearing voice you might hear like feel like you're hearing voices and the person will just report that i mean having an unusual weird distressing experience but they wouldn't go on to have a delusion because the delusion is the result of a problem at the stage of as they call belief fixation or metacognition okay and if this is really right then there should be cases in which people have the unusual experience but they don't go on to have a delusion and of course we are all familiar with this in everyday life if you think about it you've probably had uh feelings of deja vu or jamaica you know in deja vu you think i've been here before i've had this experience before even though you know that you haven't okay and this is the point that you know you can have an experience that is completely anomalous or weird don't go on to have a delusion about it okay and in this case you say well it feels as if i've been here before or in the opposite case jamie vu you know you walk into your bedroom and it feels where you sleep every night but you just have this fleeting instant of what this is new or this is feels like a duplicate of my bedroom or something but you don't go on to form a delusion because rational insight is retained unlike the classic case of delusions of misidentification which i spend no time on in the delusion of misidentification a person will say something like they'll see their wife or see their husband or boyfriend or something and say they're a stranger see and you say why do you say that you know don't they do they look

the same as their appearance and then they look identical but they are like a replica or a duplicate or an imposter in the classic delusion and the story is what has gone wrong here is there's a problem at the level of sensory processing this thought is that normally when we see a familiar face we get an affective response to that face which is part of a signal that that's a familiar person you know i see you i see my sister or my my partner and then it's like oh yeah that's that's her so it's a feeling of familiarity that's based in affect but when that goes missing then all the other perceptual information is intact i recognize the phase the shape of the face and so on but i don't they don't feel familiar and in the case of delusions of misidentification the person rationalizes these losses or alterations to the sense of familiarity by saying oh you know my wife's been replaced by an imposter okay amazing delusions but there you go but that's just an example of the kind of two-stage theory of delusion formation which was developed for delusions of misidentification now in turning to the quarter division what is the first stage within this framework well alexander bion says that it's the loss of what people call the sense of mindness or uh subjective presence is what anil seth calls this and people from sussex who work on de-personalization basic idea is that something that philosophers and many people have remarked on or thought about is that when you have experiences part of that experience is an awareness that you are the person who's having experience as jasper's said every psychic manifestation perception bodily memory it carries with it this feeling of being mine and having an eye quality of personally belonging it's been termed personalizations and so the thought is just as when you see familiar people there's this kind of subtle um just this little subtle feeling of familiarity which you don't even notice that it's there until it goes missing and so it's the case that with your everyday experience it feels like it's yours i mean who else so there's this kind of aspect of subjectivity or subjective presence that you wouldn't even become aware of until it went missing we only see it as we only even notice that this is an a kind of subtle accompaniment to all of life until we are confronted with as beyonce's exceptional and pathological cases and then if it goes missing you're really in trouble right so for example in case of pain a symbolia what can happen is that someone can have a pain they can have the experience of pain you know um and they can report that experience of pain saying yeah you know i mean yeah i'm in pain um but it doesn't feel like it's my pain or i feel detached like it's not happening to me until you know when on interrogation you say yeah yeah you know it feels hot i feel kind of pressure it feels intense in other words they can be still able to detect and represent uh bodily damage through the process of as it's called you know nociception the primary representation of pain stimuli okay but something else has gone

missing right and what uh in this nice um and there's two theories about penis symbolia one is that what's gone missing is an emotional response to pain so that would be the thought that in the normal case of pain you have no conception you know the detection and representation of bodily damage and then there's if you like an affective or an emotional response or interpretation to that and the resultant pain experience is like a merger or combination bodily damage plus uh distress and that's what you report but when you take the distress part out you just get the representation of bodily damage that's one theory of pain but colin klein had a different account and on colin klein's view it's not a loss of affect this is not a loss of affect it's a loss of the sense of mindness so a symbolia on his view is a localized version of depersonalization experience asymbolia is like a restricted case just just to the pain processing where the connection between uh the processing of bodily damage or nociception as it's called subtle feeling that the experience belongs to you comes apart so as he says this sort of unity may fail and that this leads to what he calls depersonalization or pain as opposed to depersonalization in the general case where the the loss of the sense of mindness is not just for pain but for everything so someone with severe personalization will move through life you know having sensations sensations of movement you know perceptual sensations bodily experiences but feeling as if those things don't belong to them as someone says here it feels almost like i have died but no one has thought to tell me so i'm living in a shell that i don't recognize i felt my brain was somewhere else and just watching me i couldn't tell whether i was present or whether i was the part that it was gone there was the one watching and the other it's like seeing life as if it were played in a film or a movie but in that case who is watching the film i find myself regarding existence so from beyond the tomb i'm outside my own body and individuality i'm de-personalized so just to recap okay klein gives us the idea that you know symbolia you lose this this sense of mindness or belonging to me for one particular uh channel of bodily processing that's um nociception or pain signaling but in depersonalization this loss is global and then and when the loss is global the person reports depersonalization for everything and the feeling there is like well they say it feels as if i don't exist it feels like i'm not there right and so how should we explain what's going on here well there's different theories uh beyond has one i had one 20 years ago it's a bit out of date now i think i'm updating it today even as we speak and then we have the one i'll be talking about uh today which uh we owe to the active infant skies like seth sakirus and so on but here's beyond's take on it which is very interestingly so dion uh what beyonce says is that depersonalization is this case of an anomalous experience and the anomalous or strange

experience is the feeling that the self is unreal and it's not there and so if we go back to the cotard delusion remember the two-stage theory of delusion is that a delusion arises when you have an anomalous experience and then you have some problem at a higher level of rational interpretation narrative organization or metacognition so that you uh you have no rational insight into the experience or are unable to explain it or you end up just taking it for granted and that's beyond's idea about depersonalization it's the first stage of cotard delusion so when someone with depersonalization says it feels as if i don't exist it feels as if i'm not here it feels as if experience is outside me or i'm not involved in it that's because they're having the same experience as the kotard person had the kotard patient has but they have rational insight and intact metacognition so they can look they can as it will look at that experience and it feels as if i don't exist it feels as if experience isn't mine i know that's not true so i report it just saying it feels that way but i don't they don't develop the delusion so what differentiates cotard and depersonalized delusions is that the kotard patient has the same as a um as a patient with depersonalization but they have intact rational insight um that's the philosophical angle but recently this very interesting group of guys in mexico have a neuropsychiatric version of um alexandra beyond's two-stage account which is they explain mr a's case you know the man who thought that his organs had dried up and that he was he has insular cortex dysfunction and that insular cortex dysfunction is the first factor okay so the insular cortex dysfunction produces the experience of depersonalization the feeling that that one sensory experiences and cognitions don't belong to the subject or that and then there's a prefrontal cortex dysfunction that produces the second phase the second factor so this is the story if you look on the left you've got uh the um the first stage insular cortex um produces this anomalous subjecting feeling state so if you have hypoactive insular cortex or maybe lesion or you know neurochemical disruption then you're going to have insular cortex problems and that's going to lead to the feeling that experiences are not yours in some way that could manifest as pain a symbolia depersonalization or it's also in alexa thymia which we didn't talk about today but in any case that gives rise to this abnormal sensory state or if they call it abnormal perception so something has gone wrong with your ability to feel that experiences belong to you thanks to your insular cortex dysfunction if you also have prefrontal cortical dysfunction then you can't explain that you have no rational insight into that and so you end up with the delusion now just to recap the um the assumption here that's shared by beyond and by ramirez bermudez and his collaborators is that the first stage is essentially the same or similar between the personalization experiences and qatar so the anomalous experience of depersonalization with

rational insight well that's depersonalization but if you have the anomalous experience of depersonalization with no rational insight then you get the cotard delusion right and the result and this is because of hypoactivity or lesion maybe in your but this is this is where i want to go today this is wrong right this doesn't fit with the phenomenology of cotard syndrome or the neurobasis of cotard syndrome which is typically a profound and catastrophic reconfiguration of the bodily and effective life world of the patient it's not a restricted quasi-perceptual deficit and it's not just um insular cortical um because look look at the variety of conditions in which you see cottage just some right affective disorders schizophrenia syphilis cephalo encephalitis hemorrhage malformations brain neoplasm the cancer migraine parkinson's disease dementia right this wide spread catastrophic conditions of you know neural malfunction and then and then you see um a quater delusion it's not just the ins not just the insular cortex um hypoactive okay and not only that it's not just cartilage delusion doesn't look like a i'm just like oh it feels as if i'm not here if you go back to the initial description of the symptom of symptomology it's not just i feel like i intact perception and cognition it just feels like it's it's not mine or i feel detached from it i mean typically some of qatar delusion undergoes like completely profound and uh quite catastrophic reconfiguration of their whole life experience they feel like their body is expanding or constructing or you know that guy felt as though his organs had dried up they can't sleep they stay urination and sexual function and everything goes to um crazy their appetite is out of luck you know it's a complete catastrophic breakdown uh often it's progressive but there's no doubt about it that's a catastrophic breakdown so there's a first one we can disagree with the foundational assumption but at the same token but at the same time sorry at the same point it is true that insular malfunction is um associated with cottage with with uh depersonalization i beg your pardon so let's look at that first and it's also true that um the insulin malfunction among other things is profoundly involved in cataract dilution and i think we can see why and this is where active inference makes its appearance enter stage right active inference theory uh active inference and predictive processing provide this really nice framework to um tie all these ideas together to explain depersonalization cotard asymbolia and to fit together the kind of neural correlates the phenomenology and symptomatology symptomology okay well first let's let's start with the kind of the restricted thing let's talk about uh depersonalization why would insular damage lead to depersonalization why would it be the case that if you had hypoactivity in your insular cortex you would feel like your experience wasn't happening to you it's pretty interesting and amazing uh sensation well the reason is that the it does a lot of things you know as they say it's an enigmatic um and

ubiquitously involved neural structure it's very kind of deep it's kind of it receives afferents from almost everywhere it communicates with almost every every part of the brain okay and you'll see it involved in in so many things so it's hard to it's always been very hard to work out a particular function that the insular cortex does right it's not like you know for example looking at part of the visual cortex that you know processes wavelength information or something like that or is involved in you know these neural circuits are involved in determining movement or trajectory or something like that it's not like that the insular cortex somehow seems to contribute to so many things as a as an integrative um device of integration and i would say as a device of organization and coordination and it's like a relay station actually between all these other different things but the posterior insular cortex is very important in somatosensory vestibular and motor integration so that the posterior insular cortex is basically involved i would say in integrating all the information about what's happening in your body right while the anterior insular cortex which is our real culprit here has pres projections to limbic regions which are you know that's one of the substrates of emotional processes and is associated with integrating autonomic and visceral information in other words the basic information about how your body is functioning um you know my blood pressure my blood acidification oxygenation you know state of my organs and so forth this kind of really basic keeping you alive uh kind of information integrating that with emotional cognitive and motivational functions in other words and here's the killer it brings internal signals into consciousness and emotional interpretations supporting the neural constructions of subjective feelings so um just to go let me stay here for a moment so the thought you know that we get with people like bud craig and everything like that is that the anterior insular cortex basically takes all these signals about what's happening on the inside on the inside of your body you know my blood pressure's up my heart is beating whatever my organs are functioning well or not and integrates that with all the other uh cognitive systems that are telling you what's going on outside you know what you see what you hear what you smell and the higher cognition what you remember what you mentioned what you're thinking about okay and that creates it integrates all this in other words to kind of to allow you to to create if you like a representation of how what is happening in the world is interacting with what's happening inside my body to enable me to function adaptively in context and as they say well what is that except a sense of self a sense of sense of self or representation of their self is i'm the thing that has to manipulate my body around the world to interact and engage adaptively with it and to do that i need to integrate all the information that's relevant to that task that's

what we need a sense itself for and that's what the anterior insula does anterior insulator looks down at the bodily processing looks out to the world and integrates them to give you a sense of as they say subjective feeling in the moment now so with that in place we can see why when um anterior insula goes wrong you or you know malfunctions you might get a uh the sense of not being involved in now the most um persuasive i think explanation of how this might go wrong is that it's a dissociative response to trauma this comes from medford's here and david the people who work on it but they explain it in kind of emotional terms so they call it they call a result as emotional numbing so the thought is you're having a traumatic experience you're being raped or being bashed or maybe you're in a you know destructive abusive environment as a way of coping you know your your your brain shuts down the anterior insular cortex it just turned it just turns it off and what is the result you still end up and you still end up you know seeing feeling thinking and so on but you don't have the kind of effective as i say effective response or as as uh sierra puts it you no longer have the emotional coloring of experience so it leads to a qualitative change of conscious awareness which is reported by the subject as unreal or detached so the basic idea is you shut down affective processing so that you can cope with extreme distress or trauma and to shut that down you shut off the anterior insular cortex so you no longer feel that it's happening to you so you can see what you have here is really an effective or an emotional theory of self-awareness the basic idea is self-awareness is really created by the emotional coloring of experience and when that is gone you end up in a emotionally monochrome world so to speak and there's some evidence that lack of anterior insulin insulin is related to a diminished emotional responsiveness seen in dpd and that when if the insulin comes back online um not only they restore their effective responses you know um then they start to feel that they are this feeling of detachment and disengagement disappears i just make the point that um this theory of um this theory this inhibitory theory of depersonalization is dissociative it actually it really requires a predictive processing framework because it's not just the fact for example that the subject doesn't feel effective responses it is the fact that it doesn't feel effective responses in contexts or conditions where they are predicted where they should occur you know so depersonalization arises when you know i don't know take take a mother who's just given birth and then someone gives her the child to cuddle or nurse and she doesn't feel anything right or you come home from a long trip your children run up and jump into your arms and give you a big hug you don't feel anything or you know you hear bad news um you hear terrible news someone close to you has died or whatever don't feel anything now it's not just that you don't

feel anything that's the problem it's that your mind predicts that you should your mind has this kind of ongoing model of the world and you and you in the world that your progress through the world and your interactions with the world will be as they say emotionally colored and these these emotional colorings are effective responses which get you around the world and when that's gone right it's not just that it's gone so you know because we have plenty of experiences that are emotionally neutral walking down the street you know i don't know doing your tax return um reading a book you're not especially interested in all that kind of jazz listening to a talk you know going going to a department meeting or that is a slightly distressing experience but you know you know what i mean the mere fact that there's no effective response doesn't make you feel depersonalized it's the loss of affective response in a condition where your mind predicts it it's that it's the um it's the disjunction between or the the contradiction between there should be affect and now there isn't and that's reported by then that's the thought is i feel like i'm not here okay and so the hypothesized mechanism is that the you in a traumatic situation and it's too much for your brain to deal with so these inhibitory processes kick in to shut down the affective response in the moment as a dissociative response and its inhibition of the anterior insular cortex by this inhibitory structures and this is exactly what i just said why that produces depersonalization is because affective experience is important in um [Music] in getting around life and also it's predicted you know your mind's model of you acting in the world is that you will have it have uh effective responses to the world here's where the affective inference um starts to kick in okay so then the question is how does the mind predict that your progress through the world will have or will give rise to effective experiences how does it do that well i think it does so in a process of what is called uh inactive inference framework predictive processing framework hierarchical self-modeling and basically hierarchical self-modeling is as i said before briefly that um just as the mind you know makes uh models or representations of the external world the world out there in terms of its causal structure then the mind has to make a model of what's going on inside the body in terms of its causal structure and the way the mind does this is by integrating all the properties and features of the world and attributing them to unified persisting causes well that's objects in in the case of the external world that unified persisting causes are objects you know you don't see smears of colors and movements and changes you see cars driving down the railway you don't see these kind of weird interacting shapes and properties you see people's faces smiling and so on it's kind of obligatory strategy for the mind to integrate properties and features by attributing them to underlying objects or in the case of substances if you want to call it that

unified persisting things all right and that's what that's what we do in outward and the idea of the higher of the self-modeling framework is that's what we do inward as well the mind's obligatory strategy is to unify and integrate when there are regular coherent predictable flows and sequences of experiences or sensory information the mind explains that if you like interprets it and predicts it by attributing it to the same object this happens at multiple levels right at the most basic level you know you have to integrate all these different different homeostatic uh functioning you know blood pressure uh visceral dilation you know vascular dilation all this kind of stuff that keeps you going well actually you don't that is actually integrated into by attributing all of those things as occurring in the same entity and we feel the result in what in an interceptive state interception being the internal analogue of perception so example which makes this point well i think is uh fatigue right well when you are fatigued right what do you feel you feel like i'm fatigued the the thing you know the embodied thing feels depleted but you don't feel uh all these individual streams of you know glycogen depletion adenosine depletion um slow metabolism all that you just get the kind of overall vibe of i'm tired right and that's good because then you can go to sleep you don't have to fine tune anyway you can't you can't get in there and change your glycogen levels or anything like that you can just eat so hunger is another example of a kind of interceptive sensation that is um you know this does this kind of aggregative and integrative thing this kind of type of self modeling this is very familiar from the predictive process from from that takes place at multiple levels given the kind of imperatives that we have to survive okay so let me just spend a little bit of this okay at the most within the um active inference framework okay the most basic task of any organism self-organizing system is to survive it has to maintain its um maintain its physical integrity you know its boundary with the world um it has to take in energy from the world you know it has and so on has to engage with the world take in energy from the world and then it has to do something with that energy to maintain to keep all its components interacting so that's at the most basic physical level organisms avoid this inevitable tendency to thermodynamic decay that we all face and you know the probabilistic and analog of that is they avoid entropy they have to keep themselves in a state far far from entropy by taking in energy and using it of course at the biological level what does that actually mean that means staying alive surviving prospering in and the most when we look at the physiology then obviously we're talking about homeostatic mechanisms we're talking about mechanisms of self-maintenance for a um for a discrete you know unique individual so to speak now there can be obviously cases where um borderline cases of home is homeostasis

where you know where organisms are dependent on other organisms for example there are lots of cases where there's a kind of you know criss-crossing of boundaries but let's keep it simple for the moment think of a cell or think of an organism got to maintain their their um basic functioning and that's what referred to as homeostasis or allostasis slightly larger more sophisticated concept so these are the kind of basic physical imperatives but of course to do that there has to be a lot of um signaling within the system you know the levels of levels of endocrine function levels of blood pressure levels of blood oxygenation they are signaled to a brain or a protobrain or a mind if you want to call it that and that signaling okay that signaling requires that the organism in some sense implement or maintain a model which predicts what is um what is uh optimal for it how far it's departing from optimal and whether the actions that it's taking to get back into optimal range are successful and when it can do that when it can successfully model itself then it's minimizing what they call variational free energy and surprise here just refers to departure from the predicted optimal state given the organism's action now if that sounds too abstract just think you know here's your brain um and you're riding up you're starting to ride your bike up a hill or walk upstairs okay and you have a kind of good optimal range of heartbeat function and blood oxygenation okay but that needs to change now because you're walking upstairs right you need to you need more oxygen in your blood you need to work harder and raise your heartbeat so your mind you know sends if you like brain sense instructions raise heartbeat you know this is all at the kind of signaling level it's quite dynamic if your heartbeat goes up that's good it should be up when you're going upstairs and it should be up if you're riding your bike up a hill so you've minimized you know free energy you've stayed within your optimal bound okay and you're not getting in any information that's um out of out of the kind of predicted um yeah that violates a prediction my model for that situation let's say that you come back downstairs and you sit down and you're resting and having a drink and your heartbeat is still racing well that's outside optimal range that's not what your um your body and mind you know predict for that situation your brain think your brain doesn't well your brain doesn't think but yeah now there are signals saying this is out of kilter i'm sitting on my couch my heartbeat is 200 or rather i'm sweating sweating furiously but i'm not riding my bike okay and it's not 40 degrees outside what's going on i've got a so a lot of signaling going on all the time about whether and how you are optimizing your bodily function and so that's just that the level of signaling and some of that's very very dynamic and leads to these kind of automatic adjustments but of course at a higher level when you talk about um cognition or what you can think of the mind or as what we call cognition as

the construction of sort of representational or computational systems you know that exploit these information or properties or that organize these informational and signaling properties so that we end up with a representational architecture that's how we get perception memory sensory motor control and so on and what that's what the theory of computation gives us for understanding the mind it gives us what the theory of computation basically is in its most abstract is turning information information and signaling into representation representations that can be manipulated okay translate these abstract ideas into the idea of self-modeling and we get something like this we get okay at the very basic level of organismic function then you have allostatic or homeostatic integration your mind is kind of your brain is looking around am i in or out of am i in or out of optimal functioning all this signaling is going on and what we call interception is the integration of those processes okay which attributes them to the thing which is being regulated the thing which is being managed okay the thing in which all of these um processes have to cohere to enable adaptive functioning and then we get a readout of that and the readout is your interceptive feeling fatigue you know if you feel tired that's because you know glycogen depletion adenosine levels have changed you know metabolism is slow blah blah blah muscles are weaker and and not contracting as fast and you feel that as fatigue as its integration of all of those allostatic variables and that happens basically the job of the posterior insular cortex written there as pi is to do that that's where all that stuff coheres right that's where all of that stuff coheres then of course there's not just the issue of um how is my body functioning that's one thing okay is my body de-energized well yeah that's important to know but then there's a separate kind of we might call a high level issue of the meaning and significance of those bodily processes for you in context right so if your heartbeat is racing and you're riding your bike up a hill or going for it in the gym that's a good thing right that shouldn't cause you any particular distress you'll feel that as exhaustion but if your heartbeat is racing because you're waiting for your cancer test you know and the doctor has just come back just come out of his office and he's looking at the um he's looking at your chart and he's looking at your um he's looking at your ultrasound with a look of uh concern and worry now your heart is racing right why is it racing because you're worried that you've got worried that you might have cancer that's a different thing but the point is that the feeling is different the feeling of worry uh distress anticipatory you know anxiety well that is characterized by uh racing heartbeat and and feelings of tension but that's the same bodily signature that you can get when you're powering up hill on your bike or you know or something like that so the level of heartbeat in itself is as it were you know effectively or emotionally neutral what that level

of heartbeat depends on and how it and how you should react to it well that's a product of the kind of emotional context and to understand the emotional context that's when you need to bring in the rest of your perceptual memory and so on faculties and indeed your self-representation of you know how i am functioning in the world and so to do that you need to uh interpret or i just put it here transcribe those bodily feelings into emotional feelings of sadness or anxiety or whatever they are i've given the example here of um fatigue because it comes up later but it's also a good one right if you're fatigued you know if you're just tired well that is a certain bodily feeling no doubt about but if you're depressed you know someone who's got persistent depression they also feel fatigued they feel tired lethargic apathetic unmotivated unenergetic but fatigue in an episode or sorry d energized bodily dna judge bodily de-energization in a episode of um sadness is experienced differently to just fatigue that's the point right and why is that because in an episode of sadness the world and your prospects in it have been appraised or evaluated or cognized choose your term that is because your your mind has basically determined that the world for you is a site of um irretrievable disconsolation you know there's losses in there that you can't recover or recapture and of course so um that produces the feelings of sadness okay so why does this so when we translate this into um self-modeling you've got the kind of physical level of bodily functioning and the kind of homeostatic and signaling stuff but then in order to interpret the significance of all of those changes you need a model the mind needs a model of your organism and how it's functioning in the world and what it should expect from the world and the kind of affective consequences that it should get from the world and so it makes multi-level um self-model at the most basic level at the most basic level you know the mind models itself as um as a unified entity in which all of these bodily signals cohere and then it can and use that model to regulate the body now you can go to sleep if you're tired at a higher level slightly the mind models itself as the entity that has these feelings these feelings of sin these affective feelings these feelings of sadness anxiety or these feelings of happiness pleasure resentment you know all these subtle feelings that are modulated by your actions as you go through the world so just as you need a model of the uh the bodily functioning to model the body at a basic level to control once to to get through the world well we get through the world by acting to manipulate our feelings so we need a higher level model of ourself as the subject of affective states okay and here's the and at the height it's still higher level still we report those feelings in a kind of verbal narrative way you know i'm late for work you know i'm studying i'm studying you know i'm practicing gymnastics you know i'm married etc all these things well this is just the kind of the high level

narrative self model that we use to report our feeling we used to report our states and communicate them and engage with other people and so we engage at this narrative level there's a narrative level self model that's kind of linguistic but the narrative level self model is supported by a effective or emotional level because as we engage with other people and report our states what we're doing is doing that to change our change the way we feel to feel better and at a still lower level those affective feelings what are they they are transcribed bodily feelings that is to say they are emotionally interpreted bodily feelings so the um the self model is a hierarchy is hierarchical a basic level it's it's probably at a status higher level it's bodily as interpreted for its emotional significance and at the highest level it's narrative now here's your um here's the hypothesis about um all these disorders in a cotard delusion the self model just disintegrates from the bottom up right it just falls to bits it falls to bits at the physical level right at in other words this the system becomes unregulated and unpredictable right the the mind is sending his mind is sending signals control sleep control appetite control metabolism um control movement you know get up to it but they're not getting any traction they're not getting any feedback they are the whole kind of modeling structure is disintegrated so that what the mind learns so that the mind actually can't make a model which predicts the functioning of the organism it still has this kind of sensations of organismic function but they're not integrated unified and modeled as belonging to a continuing thing because and that's precisely because that model is made for the purposes of regulation and when the mind and when the body becomes unregulated the mind learns that there is no there is no model that's going to predict and regulate its behavior so in other words if you think of the self as a model that the mind makes at the most basic level for regulatory purposes then when it's when it's completely unregulable the mind learns that there is no model that can do the job we won't say oh yeah so that's the that's my explanation of cortad okay well but in de-personalization it's not quite like that right because in de-personalization basically the model is intact okay the model is intact you've got kind of controllable bodily functioning you've got cognition perception and so on all working but the only thing is that the affective processes have been inhibited so this kind of you're seeing the world feeling the world touching and engaging with the world but there's no effective responses to the world because they've been shut down for as a dissociative response so now you've got an intact self-model but you can't annex or incorporate effective states into it so you indeed that's why depersonalization people feel as though they feel as though their bodies the same their perceptual and sensory motor states are the same and their narrative ability to report that think about that you know their higher level

metacognitive states and narrative reportage they are intact but affect is missing effect has been if you like detached or taken out of this model that's otherwise intact and that feels as if they don't think they feel detached and on the outside of experience that's why they say they report they feel as if they don't exist but they don't say they don't exist because the model still the model is basically intact just with this uh subtraction of its effective component ability now pain and that is as a result of unpredicted deactivation of the anterior insula because anterior insula is precisely the structure that integrates all the processing that does those kind of evaluative emotional processes and produces effective responses now in pain a symbolia which is reported as a kind of restricted depersonalization that is actually a problem at the lower level because pain a symbolic is a problem at the lower level of the posterior insular cortex which is more at the level of the sensory integration not the emotional interpretation so in pain a symbolism no susceptible signals are not incorporated in the body itself or incorporated in the body itself model so that's a kind of early process so once again in pain a symbolic you've got an intact self uh bodily self-model affective and narrative but somehow no susceptible signals aren't being incorporated into that model so and since they're not incorporated into the model at all then they won't be transcribed at the effective levels of self-modeling and then you'll and then the um the mind is left with well i'm experiencing no susceptible signals but i'm not they're not incorporated in the kind of bodily self model at the basic level so they're not mine so to speak i'm anthropomorphizing on behalf of your brain here and then since they're not incorporated in the bodily self model they're not going to be effectively interpret interpreted at higher levels of emotional self-modeling and once again the subject is left at this kind of narrative level just reporting on the subsequent experience it feels like the pain is not happening to me and so is this self-modeling theory well this self-modeling theory um comes from of course effective i'm sorry active inference and predictive processing theories right uh and this is um from seth and fristen you know they say that on this theory of interceptive this might slide here am i going the wrong way on this theory of that that feelings are generated at the bottom level by this kind of successive interpretation in terms of models which are ultimately top down and within this framework it's as i said the anterior insular cortex is specialized to allow us to integrate information about body state to feel its significance and the point about the self-modeling theory is those feelings of significance or effective feelings are attributed to a unified persisting entity the self okay which is the self model i'm just rehearsing repeating here what i've said and as they say there's another way to put it what i've just said emotion and embodied itself would have grounded in active inference

of those signals most likely to be me in other words as you go through the your mind needs to model the consequences of your actions and the consequences of those actions are going to be you know interceptive internal bodily and also they're going to be emotional and affect and so at any given time we have a self model which is um predicting how our actions with the world will make us feel and what the kind of sensory consequences those actions will be at all levels from homeostasis body integrity body integrity and so forth and as they say the metacognitive and narrative eye or as i've put it to simplify you've got kind of basic bodily self you've got and that gets transcribed and interpreted as a um effective or emotional self and then um at the top level we report and communicate that with the narrative self basically the self is there to serve as to represent the source of endogenous signals arising from inside and to be the target of internal regulation and this uh this goes back and it craigs papers in the early thousands you know to put it this way that you know within the posterior insula that does the basic bodily stuff you know heartbeat oxygenation energy depletion and then those representations of basic body state get progressively and progressively remapped at higher levels exploiting uh more more higher and a modal cognitive processes until we get to the anterior insula the anterior insula has to communicate with these other systems of perception memory inference imagination to tell you the experience to tell you what to tell you sorry what the uh the meaning or the significance of all these bodily changes is and should these bodily changes are they adaptive in this situation i'm running my bike uphill high heartbeat fine my heart's racing when i'm when i'm having a conversation with my wife this is bad this is kind of anxiety um and so forth okay i can skip that this is just you know to say that the anterior and posterior insula they are hubs they are not a place where this they are not the kind of there's not like a self module a bodily self module in a effective soft module it's more that they are kind of integrative hubs that basically direct traffic is a good way to put on or another way to think of it is they're like the read write head on a um on a turing machine right they're coordinating and doing stuff they're integrating stuff across them across the brain to this particular task all right and and they communicate with other hubs prefrontal and amygdala which are very important for collecting information for the evaluation you know the prefrontal cortex more about collecting information for at the country it's a conceptual level you know thinking about you know is my have i have i offended someone is that why they are you know is that why they're angry or ignoring with me is that because of that kind of thing whereas the amygdala is more like coordinating information about basic perceptually driven stuff you know someone raises their voice or puts on an angry expression and within a couple of

hundred milliseconds i feel a response to that so all right so so where are we to summarize right these are just a couple of quotations about the the insula from various things let's just read one right the convergence of multimodal sensory information and ability to read out subjective states explains why the insular is intimately involved in effective processing correct okay it's functionally important for depicting salient information and interfacing between physiology and higher order cognitive systems and that's why it is so functionally ubiquitous and connected with so many things to form a salience detection network it allows us to feel how things matter okay but don't within the predictive processing in active inference it's not just that you feel salience salience means salient to me relevant to the organism and its prospects i just something can't be salient unless it's relevant to a goal really relevant to a goal of survival or salient to a goal of you know maintaining social relationships or achieving um some ambition so salience is not something in itself salience is relative to particular goals of organismism survival or organismism prospering at various levels and in order to do that the mind needs to model itself as the bearer of the goals as the thing is prospering or not is affected by its actions now here's the thought that is ethic if you have just like you have bodily states like interception you have affective states like anxiety or depression and what they what those states do is they tell you how you're prospering they tell you how you're prospering bodily and they tell you how you're prospering emotionally which is great and the good thing about that is then you can act right you can perform your active inference of foraging in the world for solutions of moving around and sampling sensations and trying out activities and so forth all these kind of ways of engaging with the world that is going to optimize you know your ability to build a model of yourself as a functioning system but you don't actually have to um fine-tune all the kind of the mechanisms and low-level systems and that's good because we can't right so if you think about fatigue i cannot go in and change the metabolism of all my cells i can't go in and you know change adenosine levels and i can't change the balance of serotonin and acetylcholine across my brain but i can go to sleep so i manage the feeling of exhaustion and then i wake up feeling better having optimized my um optimize my organism i optimized my organism by managed my feelings and similarly at a sort of higher level of the emotional functioning i optimize my organism by managing effective or emotional feelings so you know if i feel sad i withdraw from life and and avoid the context that make me sad if i feel happy then i'll keep doing what i'm doing you know if i feel amused or interested or engaged then that'll lead me to kind of pursue and keep performing the actions that are rewarding and so on and so on and so life then becomes a matter of acting to regulate affect by active

inference as a proxy for minimizing predictive error or in other words optimizing organism function across the function of the organism so in other words and the self here is the thing if the self is modeled as the thing which sustains these feelings and the thing whose feelings can be changed by active inference or action then that's the role of the self the role of the self is to be the entity okay that we can manipulate okay so in this sense is like a proxy for overall organismic the self the feeling of being the feeling of being a unified persisting allows us to get through the world by manipulating our feelings so the self model it's like the airplane icon on a control panel on a plane right so the pilot flies that boeing 737 or 747 or whatever by manipulating the controls and you can see on his panel you know the little model of the plane banking or going up and down well that's great because the pilot has no access to the electronics the software the hydraulics and all the mechanisms right pilot can't impose not going to go back down to the tail and try and you know change the ailerons or anything like that and the pilot's not going to go in there and you know mess around with the software that of the computer that controls the system that's all delegated right that's the way down all the pilot has to do is manipulate the icon and that's all we have to do we have to just act to make ourselves feel better and the self is just the target of that regulatory actions i'll skip this now help because i think i must be finished i'm really running out of time to sum up for our case of the uh the kotar delusion and these other things well in the cotard delusion what happens is you have a version of what these theorists step on and they'll call catastrophic dysregulation so in the case of ndmr and encephalitis right cellular metabolic function the whole thing is kind of out of whack because of this pervasive inflammation across the brain and so there can't be any kind of low-level signaling that's working and at the higher more representational levels you're not getting any kind of consistent or accurate representations when it's kind of converted into computational language the the brain is disregulable the body is disregulable so the system learns the system learns as a regulatory system as a target of regulation it doesn't exist of course it's that is that's just to say the self model disappears but you can still feel your body it's just that those bodily feelings aren't integrated as into a model of something that you can change and regulate and predict and anticipate okay so that's called profit dysregulation so about that my final slide has got a typo on it so i'll change that after this but just to remind you and i'll close off now because we've been going for too long but descartes famously said that it's like hunger and thirst they teach us that we're not present in our bodies as a sailor in his ship but we are closely joined and intermingled with it so i compose a single thing with it with it you know with my body i am my body i feel like i am my body we might say within

predictive process that the experience of being the subject of interoceptive and effective experiences produces the feeling the inference that those experiences belong to a simple unified thing and the more that i succeed in controlling myself by regulating my experiences my interoceptive and effective experiences the more i learn that i am that thing you know when i eat i'm satiated when i achieve a goal i'm happy and i'm worried and i i resolve my worries but states of intractable dyshomeostasis teach us the reverse in severe cases like in the ammonium encephalitis the mind has to end up learns that there's no viable regulatory target because experience is chaotic and unpredictable so if you can't keep entropy at bay back now and keep entropy at bay at the most by having a kind of model that works then your biological functioning is messed up and you've got no physiological control all the signaling goes to pieces you can't minimize variational free energy and that is at the representational level the cell model disintegrates there's no kind of self model that predictably enables you to regulate yourself and you're just left with this kind of chaotic experience in the cotard delusion which you report with the kind of the shadow that's all you've got left is you know linguistic functioning you've just got linguistic functioning rooks okay you've just got the reporting talking so that one cotard patient famously said ah i'm just a voice and if that goes i won't be anything if my voice goes i'll be lost and i won't know where i've gone so the patients this is the patient's only response to this intractable sense of bodily effective entropy the linguistic shadow the narrative eye completely disconnected from the structure of bodily ineffective experiences that normally anchor it because the small model has fallen to pieces and i should stop there because i've been going for a long time thank you you can share the screen and we can talk a little bit it's just as long as you'd like yeah sure same button as to share and then you can cool i wrote down a ton of stuff there were some questions in the chat so thanks for the awesome talk i'll start with one of dave's questions so dave had written does your evolutionary and neurological exploration of delusion especially of the pathologies of the prefrontal cortex functional connectivity preclude the validity of psychological approaches or how do psychological and psychosocial approaches benefit from a corp incorporating these kinds of methods and findings well that's a very good question i i don't think they preclude at all right i think they're i think they're vital to your psychological approach i mean i didn't i did write a it's a chapter in a book i wrote a while about this but i don't there's a bad tendency in psychiatry in the world to think okay you've got psychological approaches you've got people's thoughts their experiences and feelings and then you the kind of things that biological psychiatry does you know molecular level interventions with receptors and serotonin and that they never they

never talk to each other right and so you have a disease model of mental illness and you have a psychosocial model of mental illness you know this but clearly that's wrong right and it's got to be wrong because if you think of um think of a just an example like astigmatism or uh you know macular degeneration that's something that you experience at a personal and experiential level as you know loss off part of your visual field if you've got a black spot in your visual field that affects how you experience the world how you get around it how you interact with other people and there's a mechanical explanation there which is in terms of you know um the visual visual cortex or the retina or whatever and the capillaries and all that in the well why why would it be the case and if someone said to you oh well look you know we could only have um like a personal level an experiential phenomenological account of uh of visual experience you'd say that was mad it's oh i want to go to the eye specialist and he knows all about you know how diabetes produces this thing because we have a mechanical thing but not only that the mechanical explanation depends on a computational explanation of how vision works these cells in the retina get bombarded by photons they regret and the mind progressively reconstructs um an image of the image of the causes of that bombardment and that's our visual experience so i can tell you why macular degeneration you know at any level at the level of the retina or visual cortex it's going to affect your experience and we think that's all in order and we think that's exactly the way to do it why wouldn't we think of that for the kotar delusion or for um or or for uh you know any psychiatric disorder anxiety or whatever it's not you've got this kind of autonomous level of um autonomous level of uh you know molecular level interventions and then you've got some phenomenological level of experience and thinking i mean one depends on the one depends on the other i just think that's like that yeah so so in my view the right what we're trying to do is is try and understand what's happening in the intervening level and that's what theories of visual processing do you know are you got artificial theories and you've got neurological theories and they come together and that's what the active inference self-modeling theory is supposed to do in these cases of pathologies of self-awareness you have a kind of experience of yourself has changed and we know that that's because something is happening in your body and how are we supposed to explain that and not just correlate the change in experience with the low level cause we want to say that's because these changes in your body and brain are organized to do this information processing kind of computational representational functions and when you get these low-level changes they compromise this functioning this computational level working with the result that you have these experiences and it goes the other way you know when someone you get

information from the world it has to filter all the way down by being represented processed somehow and you know we have to take a stand on the nature of the comp of the nature of the computation whether it's wide narrow symbolic neural network whatever you know that's ongoing project so that's my it's my thoughts on that topic thank you so the next question is from dean who wrote does this does the disassociation of leave the person not able to make a plan because in some way their predictive model has been compromised so how do these people you know do they grasp objects appropriately could they get across town on the bus what level or in what ways is their action prediction but yeah well i think you know i i think people who are very effectively flattened right and depersonalized uh i think planning is an issue planning is an issue for them um precisely because you know of the idea that you know if you're trying to decide where to go for holidays or whether to get married or whether to quit your job it's not just a sort of cold cognitive process of weighing the costs and benefits those things have to be felt right you know if you decide or am i going to go to hawaii where it's warm and pleasant or am i going to go to alaska and just get well how those you mentally rehearse those options and how that makes you feel determines the decision you know and if you decide to get divorced or get married um it's not just oh yeah you know if i get married this and this and this will happen and if i get divorced the likelihood of this it's more like that makes you feel a certain way so if your life becomes effectively flattened so that perception and cognition of all the various options for action are now effectively neutral it does make it very hard to uh to plan and engage and and in the end i think what you find in depersonalization is people just kind of go through the motions with their habitual actions you know so if you're a bus driver or a university lecturer or a husband and parent then but you feel depersonalized then you just you keep doing what you're doing but you feel like uh just the empty shell going through the motions but those people had a lot of difficulty coming up with the coherence alternative plan it's like why don't you do something else why don't you you know but but we see this actually you know even in um in everyday cases of life you know you might know that it would be good if you took up exercise but it would be you know as a objective piece of knowledge it would be great if i took up you know bike riding or if i went on a diet or if i you know whatever if i moved to cities but you just know that as a kind of you know piece of information it won't make you do it unless that somehow can click into your affective self-model it's like yeah it's me i can kind of imagine myself as you know whatever it is riding up a hill walking up a you know having some future that i can actually inhabit not just intellectually but also emotionally and personally well just one thought on that

and then i'll read a question from dave is like in a way society is a big top down like in active inference we talk a lot about nested systems and you focused on sort of the level of bodily integrity and brain body but in a sense the social structure our niche scaffolds our behavior whether it's hey it doesn't matter if you don't like broccoli you're gonna put that fork in your mouth or whether it's some other type of decision there are top down priors that actually structure our decision making and planning in the absence of precision or of a good way to plan so just really interesting there i think that's exactly right and you know um i think this is really your point exactly applies really really well to addiction there's a woman called hannah pickard who writes extensively on addiction and what she points out is that an addict can know that if they stop taking drugs it would be better for them but so what you know what they need to be able to do is project themselves into the non-addicted future where they're living engaging in the world and they're getting positive feedback from the world but she says to pick up your point that's very hard to have a model of yourself like that without social scaffolding without social structure without a kind of um positive or without a kind of positive feedback and structure from a world around you so the thought is the the self that you are is as you say your priors your predictions they're not just about internal modeling they're also about the price expectations and plans that and plans and reinforcement that you get from being immersed in a sort of social media and for addicts that's not there they can't imagine they have no model of themselves as functioning in an alternative social media so when they get out of jail or get out of rehab unfortunately they get back in their old social media they're familiar triggering situations in the familiar social environment and that kind of activates their self-model as someone who who does this does that and there's and and she says when you ask them to reflect on what it's like as a non-addict imagine they can't so there's a kind she said it induces in them a kind of existential terror because um they really can't imagine themselves as a non non-addicted life it's kind of it's an empty future for them partly because she says there's no viable social scaffolding to support their their life in that world so your point is i think plays really well into that that literature and what you said there um what is it like to be you know in that case not addicted to a substance or a behavior it's kind of a subtle allusion to the famous philosophical tradition of what is it like to be a bat to be an ant colony et cetera and when you brought up i think even at the beginning you said what is it like to feel you don't exist this idea of negative space and of negation came up in such interesting ways in the talk like if you ask somebody what does it feel like to exist it's like that's 99 of the discourse or you know that's the mystery but then by drawing our attention well what would it feel like counter

factual importance for active inference what would it feel like to not exist as like i know i'm not feeling that so what is that one and then what's the other side of the coin of feeling like i don't exist it's like it's like feeling like i am in control like i am steering the ship so that was so such a cool connection that's that yeah i think that's the point i wanted i wanted to to get at you know what why i like the active inference framework for this is because the feeling of being me you know a lot of people tell you oh well if you have a perceptual experience then someone's having it you know an experience has to have an experience or something like that or how can you not feel like you're me if you have a body you can see your body touch your body and feel it but that's not what we're talking about the feeling of being me is the feeling of being in charge the feeling of being the thing that i can manipulate in order to bring about all these right and you can still have experiences right because plenty of experiences but if you lose the feeling that those experiences are things that you can change and manipulate dependent on your intentions then you lose the experience of being you and that's at a high level but the same thing imagine the thing going right down you know really low your your hypothalamus is sending signals to raise your heartbeat and not getting any your heartbeat stays low stays obstinately low or obstinately high it's the same thing i'm not i'm not getting any traction obviously a hypothalamus doesn't think that but or you know imagine a bacterium that's trying to contract its membranes to to turn left to pursue a sugar gradient and nothing's happening it's getting keeping getting this kind of obstinate feeling that i'm not in charge anymore and i've got no model that i've got no model on myself that's working to to to give me the feeling of being me and that's in such a case the system still has sensory experience sort of activity at its receptors and stuff is happening but it's just not a regulatory target anymore and and that makes me think about um when you talked about the people who are suffering from pain but they're disassociated from it it feels like it's not their pain it's a little word play so it's not a philosophical argument but gnosis and nociception it's like there's the knowledge of pain but actually it's almost separable from the suffering and so this is an extreme case where the gnosis of nociception has been entirely separated from the narrative of the self it's like yeah there is pain in the arm but that's not me or it's not part of my narrative and so it doesn't bother the person psychologically that's exactly right there's damage there's damage to the body that i inhabit is that's what the brain is saying by this body that i live in is damaged but it's not my damage now do they ever ask who's is it or does it just not seem to be like oh it's just like oh someone's car someone's parked outside or is it like i'm kidding no no um the really interesting thing i mean the early cases the really interesting thing about um symbolics is that you know

they don't take any uh action to avoid pain you know so that in in that sense you know that's really bad and so but it's it's not quite the same as um congenital insensitivity to pain which is a problem with the receptor level right those people don't detect pain and so um you know they're in a bad way they put their hand on a burning stove they um break their legs and they don't they just don't feel they don't even have the bodily sensation but pain asymbolics they have the bodily sensation but they don't annex it to themselves well it points to different um roles for different receptors so i want to ask about the nmda receptor which you kind of brought up in the beginning what does the nmda receptor do neurologically or why is it associated with that phenotype right well now you're um now you're now now you know now i'm up now i'm out of my league um but uh it i think that that kind of encephalitis is a um inflammatory thing right it's an inflammatory uh response when it hits those receptors i i think it's just because uh ndma is like it's so widespread throughout the brain i think it's like uh i'm talking out of turn here right so it's not it's not my my field so cancel this but i think it's like like a like a catalyst or modulator for so many metabolic functions right it's involved in everything as a kind of very fundamental thing so when you mess with it you pretty much you that's why you get this kind of dysregulation across the brain ah yeah that's that's the thought it's one of these it's like um adenosine or something like that you know that these uh metabolism depends on that and if you mess that up or uh what's another one uh the ketones you know if you if you get a buildup of uh phenylketone in the brain you know it's this tiny chemical but if you tiny levels of this chemical can just destroy everything produce mental retardation mutism motor disorders it's a catastrophe an absolute catastrophe because the cellular level functioning is so dependent on these nano nano level adjustments of these basic things speaks to two dimensions of the challenge which is the multi-scale systems there's little things like molecules and big things like people and then also the trans disciplinarity and it's like we should have one story for one level from one discipline and then a separate discipline and a separate story we need something like an integrated approach what's that going to be that's exactly right and so where i suppose where i came into this when i was you know when i got into philosophy and people thought well that what's the integrative story going to be well you know some kind of computational story top-down classical computational modeling cpu modular inputs modular outputs that's one way it's a kind of broad brush approach then you have neural network stories and then you have you know deep learning stories and all these different architectures and now you have and they have active inference and it depends what you think uh active inference is gonna um how it's gonna play out i mean if if you think of

active inference as you know at the biophysical level it sounds right but it's very abstract right it treats like a human organism as like a physical system in space it's like a self-organizing asteroid or something like that is really the same the same laws of physics apply and then you have the kind of bayesian implementations of it's an interesting question you know whether you think the brain is actually bayesian you know because a lot of psychologists will tell you that the brain departs from bayesian optimal reasoning almost everywhere in almost all and almost always you know so yeah very interesting and welcome back dave so um one of the questions from dave um was have you looked at functional delusions socialized as in mobs or depth psychological no no i'm just i'm still on the um i'm still on the standard curriculum of um delusions of misidentification cotard delusions schizophrenic schizophrenic delusions i haven't talked about maths psychology and depth psychology i had some thoughts but i mean clearly like self-modeling applies right uh if you're talking about how ideas proper propagate in populations then i guess you could say those those ideas which are most consistent with a self model they're going to be more easily absorbed stable stabilized you know within the individual so it's going to be very hard to get traction for an idea or an ideology that doesn't fit with a self model and probably self models are since they're constructed to regulate affective states that tells you that the ideas which propagate and stabilize are going to be those which get traction with people's um effective self-model you know what that make them feel a certain way rather than a kind of abstract intellectual argument you're not going to persuade an anti-vaxxer by giving them statistics and you're not going to get a kind of propagation of a wave of um something across social media on the basis of you know an irrational inference but but that's i don't have anything super much to say about that very true that we see sharing and activation patterns in the sort of global brain based upon affect not veracity or accuracy or any other feature um and and maybe that speaks to another one of dave's questions which was how does your work illuminate the notion of cognitive dissonance or consonants even in his technical work leon festinger seems not to succeed in building a credible dynamics of compatibility and incompatibility among psychic atoms relevant passages of gregory bateson's essays in steps towards an ecology of mind seems similarly limited so whether or not you've read those specific texts like i haven't read those specific things i suppose one way of thinking of it is within the kind of active inference or within the um predictive processing framework is you know if you get some discrepant that uh contradicts yours contradicts a self-model or contradicts a predictive model you have you know some choices you can revise the model that's you know perceptual inference as they say or you can explore the world and try and get

some you get some further information to confirm your model you that's that seems to be the active inference approach and obviously life is a balance right life is a balance between the two keep your model or get some more but it seems like if you if we take the scientific uh analogy of you know hypothetical inference often what people do with discrepant inference in sorry discrepant information is just compartmentalize it you know they just put it away and scientists of science you know known through it no one throws out a good scientific theory just because of some discrepant evidence even if that can't be reconciled you know right now no one's chucking out the theory of evolution because someone came up with a wacky organism or a wacky function you know it's like hmm there must have got to be some explanation i don't know what it is but let's just we'll just keep doing our thing and hope that eventually it'll be realized you know so i don't think people these ideas of kind of um updating or upgrading hypotheses in the face of flux of new information that's actually not true because we can't be that sensitive to information that would be mad and i guess if you have a certain self model that's getting around the world and functioning you know if you're a religion if you're a highly religious person or something like that and you live in a religious community and you go to college and you know read dawkins or something like that well i mean you have a choice you can throw out your whole self model in your whole worldview or you might just think like many surgeons scientists and otherwise rational people okay this is my life in the hospital where i'm a neuroscientist or whatever and this is my life at home where i'm a preacher and a patient family patriarch of a religious community because complete consistency across all the levels and all the functions too much but i would say because it threatens a kind of self-model it's not because it's um you know that kind of compartmentalization is not so much a matter of intellectual consistency so much as consistency with the self model and very importantly that the self model includes the sensory evidence and that meta model so even though my background is in evolutionary biology very early i realized that people who believe in evolution people who believe in creation every other flavor look at a tree and see evidence for their world model so that's the starting point for the discussion not that you see the evidence but that everybody sees evidence confirming their worldview and that's a very different discussion than the dogmatists who might be on any side on the other side right and yeah and no one is going to come no one's going to come to the top no one's going to come to the topic with no model that's going to say put aside your preconceptions well there's no such thing right so uh yeah yes definitely and that is relevant for the uh cognitive bias and ai bias discussion in these future systems but we'll have an

unbiased system what is that even going to mean yeah exactly and that is the you know the strength weakness of the bayesian thing isn't it because the bayesian idea is you know you start ab initio and then you get some evidence and you update your model you get some more evidence and you update your model and it's on but you've got to start from somewhere and so much of the action in bayesian theory is in the priors but there's no theory internal to base which will tell you where your prior assumptions about how much weight to give incoming evidence is going to come from it's a little bit more like blockchain with the way that we can make our priors explicit you don't step around this issue by using frequentist statistics or by casting tea leaves so at least we can have a discussion hey i use different priors i got a different answer versus we use radically different families of priors and hyper priors and we're converging on a similar action so maybe that speaks to at least this action being appropriate if not it being the ultimate yeah that's a that's a yeah that's a very good point it's not as though um yeah it's not as though anyone else has a solution to uh to where do we start from you know what what are our foundation what are our foundational assumptions um yeah so it's a bit rude to criticize uh bayesians for saying there's there has to be a lot of action in the prize right there has to be a lot of information in the prize to get the project off the ground true but you know they're the same with anyone you know chops get chomp skins or anyone they just yeah exactly so one other um phrase or piece that came to mind was almost like parametric hysteresis the idea that although most sensory evidence is sort of just sloughed off kind of like teflon on our mind you look away it's just totally gone there are traumatic experiences and then on the other side there are um healing modalities whether it's a healing experience or a biochemical intervention there are sort of like there's interventions restructure our models and we can think about that in active inference perspective so i just wondered like philosophically or from any view how do we think about these kinds of changing events whether kind of changing in a negative way like a trauma or in a positive way like healing how do we scaffold for that well that is actually a really good question because you know in this in the case of psychiatry it seems like um almost all therapies are equally good and equally bad like none of them really none of them really work in a way that we can kind of predict or explain there's only a statistical you know statistical correlations some people took cer antidepressants and you know certain proportion get better but then they relapse some people went to a therapist or cognitive cbt some get better some relapse you know so what kind of what kind of interventions work is really uh interesting and the kind of restructuring you know and that's the um that's part of the motivation i think for uh psychedelic uh approaches to psychiatry

where it's like okay one thought might be look i can't we can't really have a successful intervention because that would require you to rebuild or remodel yourself it's not just changing your perception and changing your experience it's the way you incorporate that in your whole kind of regulatory framework how we're going to do that well maybe with 10 years of therapy and you know lots of um patient reconstruction or or sometimes a life changing event people give up drugs because you know for some reason it's a life-changing event or they just grow out of it people say but it's hard too hard to systematize that isn't it it was psychedelics what people are saying is well let's disrupt their self model let's kind of for a while let's introduce maximum entropy into the system again they experience life briefly hopefully safely with a lot of their self-model um disintegrated yep and so they get to see life for once unfiltered you know normally everything is filtered how does this matter to me in relation to me in relation to my history my plans you know the other people that matter to all of life is you know just being filtered at every level like that and that's very hard to to experience life without that filter psychedelics give you a um a brief immersive very destabilizing experience of life unfiltered and maybe from that can be a starting point for something that's good classic william blake doors of perception um idea and also the the rebus work by fristen and and carhartt harris as well as adam saffron and actually asking like maybe there's certain aspects that these neurochemical interventions they strengthen maybe it's not just this across-the-board weakening of our world model but there are certain levels of analysis in which things might be strengthened and right right on this note though of the the treatment and the recovery one of the very early slides i think it was mr a um was the patient and then um came in initially presenting some phenotypes and then there was plasma exchange and it said after a few plasma exchanges you know he had improved and then he's had this other um thing and so i thought interesting treatments what does it say about the body and the brain or what made those doctors go for the plasma exchange and was it with with a matter right well because um because they had this idea that uh they had they had this they had this idea that inflammation you know was the core of a lot of these catastrophic the core of a lot of the symptoms when you when you look at depression and you look at psychiatric disorders people often get wrapped up in the um sensory experience and the um and the cognitive experience you know people say things that are so crazy and the thing is why would you say that why are they so irrational you know and people start at that level and they talk to them and they say try and understand their life experience and all this or whatever but the view of these uh guys in mexico was look you know look look at the other symptoms look what's happening you know

the guy is peeing his pants himself hasn't been to sleep for three days and then he sleeps for two weeks and he doesn't want to eat anything he hasn't eaten for a week you know so this is like really basic it's really basic functioning is out of control now um and that looks like it maybe that's a kind of and we know that uh you know inflammation does affect homeostasis really badly so that was their thought you know let's and then let's test them for all these antibodies for encephalitis and they find it so plasma exchange is the um the treatment for that and then of course you might think oh well all right but that's not something you're going to repair straight away you know when you've gone from homeostatic catastrophe to uh a bit of reintegration you know you're still going to be left with the cognitive level has to get back in touch with you know if you're thinking of those kind of levels of organization and systematic functioning and interaction well okay you change the molecular level function you don't change everything straight away which is why um it's a very nice study by Katherine Harmer you know give people antidepressants or anything like they give people antidepressants or you've changed their serotonin levels within an hour right so you brought their serotonin level across the brain back to now why don't they feel better why don't they change their pattern of thinking it does affect it does affect some very kind of low level perceptual processing though so they're patterns of gaze you know depressive people their circads are very avoidant you know if you show them emotional faces or something they don't look at the nose the eyes the mouth their eyes automatically avert and people with anxiety and so on so they're not getting any positive uh you know emotional feedback from their social but if you give people antidepressants that pattern of circads changes within a few hours so it's almost like you're giving them the ability now to um do some active inference and reacquire some some new information about about their world so if they go around the world luckily and they people smile at them and laugh and talk with them then they they're getting some new information but that's not going to disrupt straight away all the high level models that they've built to um you know you're not going to change their that's not going to happen straight away and that um story about the serotonin story i've personally felt is that disciplinary hierarchy it's like well we know how this drug works look at my simulation of how it binds to the receptor you bring up but then if it did that and that's simply how it worked if it weren't just um you know changing a complex biological system then wouldn't the time course of action be very different and exactly it's a it's a um looking where the light is because we can make these simulations with a docking of a molecule but as yet we lack some of these layers in between absolutely the vibrating molecule

and our own phenomenology which is sort of that fog of uh experience that we're exploring together yeah i know i i absolutely 100 agree and i think um you know yeah but most of my practicing psychiatrists colleagues and you know collaborators they would agree too but there's a big uh pharmaceutical industrial complex you know that has that has a interest in thinking you know persuading people otherwise and also people are desperate yeah yeah can't wait till we have big prior and big precision yeah yeah exactly you you brought up the um australian philosopher colin klein who i was familiar with because of the collaboration with andy barron actually about basil experience in honeybees and in insects so just was an interesting connection with thinking about okay if we're going to take this sort of bottom-up computational approach then where we take that in not just the clinical but in the evolutionary context yeah no he's a great he's a great uh philosopher for sure and he's really got his finger on the pulse of all the um i think cross-disciplinary yeah material that you need to do that kind of thing well he's really good like here's another just in these last few minutes a question from the chat cb wrote any thoughts about the role of non-synaptic information transmission across the brain especially emphatic coupling thank you kindly honestly [Applause] i don't really know um yeah i don't because uh clearly it's got to have there's no reason within say active inference framework or actually any computational framework to think that um you know the the fundamental mechanism has to be synaptic although it could it could be anything right how the brain is organizing uh the body to get through the world to survive could be anything it could help itself to any mechanism you know astrocytes glial cells whatever the hell but we have got we bought into this idea that it's uh synaptic probably going back to heb and people like that and a neural network theory is that it's a synaptic theory right you've got these units uh interconnected units with weights which are supposed to model some abstract levels synaptic functioning we might be off we might be off um off track there so maybe this is a kind of a good advantage good advantage of active inference it's like at the moment uh computational theory is hung up on the synaptic model because that's what deep learning and neural networks are right setting of setting weights across the brain but that's a very abstract idea of um heavy and you know it's a very abstract idea of what ed was on he was on about and most neuroscientists like i had a guy talking in my class the other day invited him to give a lecture he works on dragonfly vision and he was just talking about the neurons in in the eyes of dragonflies he was saying there is all this stuff in there that we don't know glial cells astrocytes what the hell that you know what they're doing what they're encoding how they're encoding it but it says if you mess with their function then you mess with the ability of the

dragonflies to do their incredibly accurate flying and landing and sensory representation so yeah so maybe the synaptic model is um once again it's a case of looking where the light is you know artificial neural networks they're great they can do a lot of things um but they are a kind of high level abstraction then again bayesian models are an even higher level abstraction if you if you want to put it that way because the bayesian mod the bayesian models just say okay you've got a system interacting with its environment and i've got a function a function that tells you how the system's going to interact with its environment and i can devolve that into a lot of if you like i can devolve that into a lot of subsidiary levels of of bayesian bayesian inference all the way down but how do they how do they glom how do they glom onto the actual molecular soup i don't know but i mean that's the that's the job of it isn't you know that's you know that's why i i guess still uh dopaminergic uh functioning in the brain is still a bit of a triumph sorry understanding the dopamine system is because you have a system whose properties map pretty well to the predictions of formal theory you know of temporal difference learning and reward learning not perfectly but it does fit so that's good so that's good you know that's good it seemed to seem to say we're in the ballpark a paper on the dopamine signaling about explanatory pluralism from a philosophical perspective about the way in which the dopaminergic signaling um in in different ways can be seen as consistent with the precision account the reward account or other accounts and so it's like maybe let's um you know wait a few decades yeah before we um you know give out the trophies let's yeah yeah yeah absolutely absolutely so i'll just very good yeah because that's right why why can't it doesn't fristen have a paper on that even you know and actually there are a couple of good papers on that now yeah with the within the active inference framework we can conceptualize the um the dopamine system as you know fine-tuning fine-tuning patterns of rather than back propagating a reward signal exactly yeah so i'll just close with this sort of um this diptych of questions a little high road in a low road one of the paths would be what research directions do you think are exciting for people to pursue and the other side is um how do you think for potentially non-researchers what you've spoken to today might change the way that they think or act i think for researchers i think i think any researchers should still keep researching you know what they're like if you've got a fascination or an interest there's dragonfly vision dopaminergic signaling whatever i think you should just keep going for it and incorporate uh active inference or any framework that that is is helping you know i don't think jumping on a bandwagon or any researcher helpful you've got a bit you've got to bear in mind that if you're trying to function in an institutional environment you're going to be on some bandwagon right

you're going to be in some lab or some in a school of thought but i think you still should you've still got to do what you're interested in rather than promote someone else's program so i don't have a i would love to see i would love to see that um that you know kind of fristonian ideas and everything could be put in a intuitive way that made them more accessible for people who were looking for a kind of a kind of recipe or something like because they're so technical and they're so formal and uh it's hard to extract um how to extract the gold you know if you know what i mean it's it and that's why yeah so so a good suite of examples would be really helpful like you know standard cases that people use to explain it would be good you know just as you know often these examples the examples are wrong right so you think of mars theory of vision okay probably wrong but at least it kind of gave you a way to understand you know what's going on in the visual system and then you can fine-tune it and change it and argue with it and everything like that but he gave people a way of taking the understanding of perception out of this out of you know out of a really kind of technical specialized framework and now philosophers talk about it psychologists it's probably become a little bit of a um an albatross around the neck of because but no one's come up with no one's come up with like a that one it would be nice if um someone could i mean because the ma framework is so um so so clear right and and it works for active inference too you can say well look the mind is in the business is what holy says of um reconstructing the causal structure of the world you know inferring the causal structure of the world outside beyond its boundaries so it's very internalist view but that's a very um plausible way of doing it you know i think yeah i get it you know you can explain that to students and to people look the brain is stuck inside its skull it can't it can't engage with the world it can only ever do it by proxy indirectly and what does it have photons sensory receptors it's got to build the picture okay and you know that's what ma said and and how he put it in the predictive processing something like that would be helpful if you get that idea across you can do it for other modalities of course interoception you know whatever sorry what was the the other question oh on the non-research frontier is um just in the day-to-day how does this influence how we might think or act just in our day-to-day experiences i think it makes a big difference if you think of yourself as um you think of your behavior as uh confirming a model you know i think it makes a really really big difference you know if you're arguing like you're arguing with your partner or you know stuck in a way of life or ways of especially especially social interaction you can think of social interactions you know i've got a thought i've got to try and get a point across or i've got a goal i want to achieve the goal right i want to make friends with this person i want to do this i want to do and then

you can get pissed off or whatever if you're if it's working or not but maybe the right way to think of it is you know i've got a model of how i function in and uh you know my brain has built this picture of how i function in the world and i'm just now i'm bungling around the world trying to optimize it you know so what i think what i think of as this kind of you know autonomous engagements of this um you individual who's above it all and choosing and whatever and you know making decisions and not really maybe what i'm doing is bungling around the world trying to optimize my model and um maybe i've got i can get stuck in a local minima and i can get stuck in a you know all my models my model's not working we had this discussion maybe i don't know if you know this philosopher dominic murphy he's a fantastic philosopher of uh science uh science and psychology but we would think of you know this national model and you know a lot of the time what you're doing is uh brand management you know yourself who you are is this model that you've made that works for you in your environment your family as you're growing up or high school or whatever like this and you've got this brand now and unfortunately or fortunately that's how other people see you your actions are really about getting your brand out there brand new brand management and if you identify with your brand great you know but if the brand if your brand isn't working for you and if you and if you just see your actions as oh god i am just what i'm doing is kind of finessing and thought and reinforcing and fortifying this model of how i act in the world does it have to be like that very nice point that instead of um social interactions maximizing my reward your reward our shared reward zero sum we can actually think about well what if we were both reducing our uncertainty about a positive social interaction what might be known as win-win in the reward paradigm but here yeah that's a really good way to it's a really good way to think of it that's a really good way yeah exactly thank you for this awesome visit it was really uh fascinating talk and one of in the chat brilliant practical advice thanks daniel and thanks for your guest thanks a lot from our lab and you and anyone else are always welcome to join anytime thank you very much it was a really great experience nice to try these ideas out and this discussion was really helpful and informed you know so thank you very much for the chance and i'll be um propagating you know i'll be spreading the news about the lab so thank you thanks a lot thanks a lot bye see you everyone bye nice talk pretty cool